

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 — SUB-STRAND 1.3 NUTRITION IN ANIMALS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use everything you learned from the Ugali and Sukuma Wiki Puzzle: your food journey maps, the Mandazi Moment, the enzyme charts, the food tests, the structure diagrams and your 3D model. Write your answers in the space under each prompt. Use full sentences and scientific words such as enzyme, substrate, mechanical digestion, chemical digestion, villi and sphincter. Always link your ideas to the school lunch of ugali, sukuma wiki and nyama choma. Where a question asks you to use evidence, say which test or observation you are using.

Part 1: Two ways of breaking down the meal

Think of the Kenyan school lunch of ugali, sukuma wiki and nyama choma. (a) Explain the difference between mechanical digestion and chemical digestion. (b) Give one example of each from the mouth or the stomach. (c) During the Mandazi Moment you chewed a piece of mandazi for 60 seconds. Describe what you noticed about its taste and explain what this tells you about what happens to the ugali in your mouth. (d) Would you expect the nyama choma to change taste in the mouth in the same way? Explain your answer.

(a) Mechanical digestion is the physical breakdown of food into smaller pieces. It changes the size and shape of the food but not its chemical identity. Chemical digestion is the breakdown of food molecules by enzymes into simpler substances. (b) Mechanical: teeth chewing food, the tongue mixing it, or stomach muscles churning it. Chemical: salivary amylase in the mouth acting on starch, or pepsin with hydrochloric acid in the stomach acting on protein. (c) The mandazi changed from bland to slightly sweet as I chewed. Mandazi is starchy like ugali. Saliva contains salivary amylase, which converts starch into sugar (maltose), so the sweet taste is direct evidence that a chemical change begins in the mouth. The ugali begins to be digested chemically as soon as it is chewed. (d) No. Nyama choma is mostly protein. Each enzyme is specific to one type of substrate, and salivary amylase acts on starch, not protein, so no sweet taste develops. Protein digestion begins later, in the stomach, with pepsin.

Part 2: How does the body know which chemical to release?

One of the puzzles in Lesson 1 was how the body seems to 'know' which digestive chemical

(a) The body does not release one general chemical. Each enzyme is made by a particular gland or organ and is released in a particular region of the alimentary canal, and each enzyme works only on its own substrate. So the right chemical is present where its substrate arrives. For example, salivary amylase cannot digest protein and pepsin cannot digest starch. (b) Starch in ugali:

to use at each stage. (a) Explain how the digestive system manages to have the right chemical in the right place. (b) Name the enzyme that acts on each of these in the school lunch and the region where it acts: starch in ugali, protein in nyama choma, and fat in cooking oil. (c) Look at your food test results from Lessons 4 and 5. Name the test and the colour you saw for at least two nutrients, and say what these results tell you about the nutrients in the meal.	salivary amylase in the mouth, then pancreatic amylase in the small intestine. Protein in nyama choma: pepsin in the stomach (activated by hydrochloric acid), then trypsin and peptidases in the small intestine. Fat in cooking oil: lipase in the small intestine, after bile (from the liver, stored in the gall bladder) has emulsified the fat. (c) Ugali turned blue-black with iodine, showing starch. Glucose and ripe banana turned brick-red with Benedict's solution, showing reducing sugars. Nyama choma and milk turned violet with the Biuret test, showing protein. Cooking oil and butter gave a milky emulsion with the ethanol emulsion test, showing fat. These results show that the meal contains chemically different nutrients: starch, reducing sugars, protein and fat. Since each enzyme acts only on its own substrate, different nutrients need different enzymes.
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Part 3: Why does digestion take different times?

Another puzzle was why ugali seems to be digested faster than nyama choma. (a) Using what you learned about the regions of the alimentary canal, compare how and where the starch in ugali and the protein in nyama choma are digested. (b) The cooking oil and butter in the meal contain fat. Explain why fat digestion needs an extra step.	(a) Starch digestion starts in the mouth, because salivary amylase is already in the saliva, and it is completed in the small intestine by pancreatic amylase, giving reducing sugars such as glucose. Protein digestion does not start in the mouth. It begins in the stomach, where hydrochloric acid activates pepsinogen into pepsin, and it is continued in the small intestine by trypsin and peptidases. Protein digestion therefore starts later and involves more sequential steps, so it takes longer than starch digestion. (b) Fat does not mix with the watery digestive juices. In the small intestine, bile salts from the liver (stored in the gall bladder) must first emulsify the fat into tiny droplets. Only then can lipase digest it chemically. Bile is an emulsifier, not an enzyme. This extra step, and the fact that it happens only in the small intestine, means fat digestion happens later in the canal than starch digestion.
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Part 4: Structures that make absorption and protection possible

(a) The small intestine is a tube only about 3 cm wide, yet it absorbs the nutrients from the whole meal. Explain how its structure makes this possible. Name the nutrients from ugali and nyama choma that are absorbed. (b) Explain how the stomach avoids digesting itself even though it contains acid	(a) The lining of the small intestine has villi and microvilli (the brush border). These finger-like projections increase the surface area for absorption by up to 600-fold without making the tube wider. Villi contain capillaries and lacteals that carry absorbed nutrients away. Glucose from the starch in ugali, amino acids from the protein in nyama choma and fatty acids from fats are absorbed into the bloodstream through them. The 'machinery' is microscopic and folded into the wall. (b) Goblet cells in the stomach wall make a layer of mucus that protects it. Also, pepsin is released as inactive pepsinogen and is only activated by hydrochloric acid in the stomach cavity. These two adaptations prevent the stomach from digesting its own protein walls. (c) The cardiac sphincter closes the entrance to the stomach and prevents acid from moving back up into the oesophagus. The pyloric sphincter controls how fast partly digested food (chyme) passes from the stomach into the small intestine, giving the enzymes time to work.
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and pepsin. (c) Describe the job of the cardiac sphincter and of the pyloric sphincter.	
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Part 5: Answering the driving question	
Driving question: How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail? (a) Write a connected explanation that follows ugali, sukuma wiki and nyama choma through the alimentary canal. Include at least one mechanical process, at least three enzymes with their regions, and one structural adaptation. (b) Choose ONE adaptation from this unit and predict what would happen to the digestion or absorption of the meal if it failed. (c) In the Benedict's test on the simulated digested-ugali sample (a glucose solution), what did you observe? Explain what this result tells you about reducing sugars and digestion, and say how it relates to what happens to ugali starch in the body. (d) Compare your Lesson 1 food journey map with your 3D model. Name two things you can now explain that you could not explain at the start.	(a) In the mouth, teeth and tongue break the meal into smaller pieces (mechanical digestion) and salivary amylase starts to change the starch in ugali into sugar. In the stomach, muscles churn the food, and hydrochloric acid activates pepsinogen into pepsin, which begins digesting the protein in nyama choma. The pyloric sphincter lets small amounts of chyme into the small intestine. There, pancreatic amylase finishes starch digestion, trypsin and peptidases finish protein digestion, and bile emulsifies the fat so that lipase can digest it. The digested nutrients (glucose, amino acids, fatty acids) are absorbed through the villi and microvilli into the blood. Fibre from sukuma wiki is not digested into absorbable monomers. (b) Example: if the mucus lining of the stomach failed, the acid and pepsin would damage the stomach wall, because the protein of the wall would no longer be protected. Another example: if the villi were damaged, the surface area would fall, so less glucose, amino acids and fatty acids would be absorbed and the body would get less energy and fewer building materials from the meal. (c) A brick-red/orange colour appeared when Benedict's solution was heated with the glucose sample in a water bath. This shows that reducing sugar was present. Reducing sugars such as glucose are the kind of product that amylase action on starch produces, so the result represents the end product of ugali starch digestion that the body absorbs. (d) At the start my map showed food travelling through a simple tube. Now I can explain that enzymes are specific to substrates and are released in particular regions, and that villi increase the surface area of the small intestine. I can also explain why the stomach does not digest itself, because of mucus and inactive pepsinogen.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Mechanical and chemical	Clearly distinguishes the two types with	Distinguishes the two types with correct	Mixes up the two types or gives no

digestion	correct examples and uses the Mandazi Moment evidence to explain amylase action on starch.	examples but gives only a weak link to the Mandazi evidence.	examples or evidence.
Enzymes, specificity and food tests	Matches each enzyme to its substrate and region accurately, explains specificity, and uses named food tests with correct colours as evidence.	Matches most enzymes to substrates and regions and mentions some food tests.	Names few enzymes correctly or does not link enzymes to substrates or tests.
Structural adaptations and self-protection	Explains villi and microvilli, mucus and pepsinogen, and both sphincters accurately, linking each structure to its function in the meal.	Explains most structures and functions with minor gaps.	Names structures without explaining their functions, or confuses them.
Connected explanation of the driving question	Gives a sequenced explanation of the meal through the canal and a reasoned prediction of what happens when an adaptation fails.	Gives a mostly sequenced explanation and a simple prediction.	Gives a list of facts without sequence or without a prediction.
Use of evidence and Kenyan context	Consistently uses the ugali, sukuma wiki and nyama choma meal and practical observations to support claims, with correct scientific vocabulary.	Uses the meal and some observations, with mostly correct vocabulary.	Makes general claims with little reference to the meal or evidence.

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